Deep Learning and Its Application

Week 4: Practicals

1. Implement the following Mean Squared Error (MSE) loss function in Python

$$MSE = \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{m}$$

Given the true values y_i = [0.000, 0.166, 0.333], and predicted values $\hat{y}_i$ = [0.000, 0.254, 0.998], use the implemented function to calculate error.

2. Implement the following Mean Absolute Error (MAE) loss function in Python

$$MAE = \frac{\sum_{i=1}^{m}|y_i - \hat{y}_i|}{m}$$

Given the true values y_i = [0.000, 0.166, 0.333], and predicted values $\hat{y}_i$ = [0.000, 0.254, 0.998], use the implemented function to calculate error.

3. Implement the following Cross Entropy loss function in Python

$$Cross\ Entropy = -\sum\sum_{i=1}^{k} y_i \log \hat{y}_i$$

Given the true values y = [[0,0,0,1], [0,0,0,1]],
predicted values $\hat{y}$ = [[0.25,0.25,0.25,0.25], [0.01,0.01,0.01,0.96]] and k is the number of classes, use the implemented function to calculate error.

4. Given the following fragment code

```python
from numpy import argmax
from pandas import read_csv
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# load the dataset
path = 'https://raw.githubusercontent.com/jbrownlee/Datasets/master/iris.csv'
df = read_csv(path, header=None)
# split into input and output columns
X, y = df.values[:, :-1], df.values[:, -1]
# ensure all data are floating point values
X = X.astype('float32')
# encode strings to integer
y = LabelEncoder().fit_transform(y)
# split into train and test datasets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33)
print(X_train.shape, X_test.shape, y_train.shape, y_test.shape)
# determine the number of input features
```

```
n_features = X_train.shape[1]
.
.
.

row2 = np.array([[5.1,3.5,1.4,0.2]])

yhat = model.predict([row2])
print('Predicted: %s (class=%d)' % (yhat, argmax(yhat)))
```

Use Functional model to develop a multiclass classification model using Iris dataset, display

a) Training accuracy
b) Prediction results

5. Given the following fragment code

```
from numpy import sqrt
from pandas import read_csv
from sklearn.model_selection import train_test_split
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# load the dataset
path =
'https://raw.githubusercontent.com/jbrownlee/Datasets/master/housing.csv'
df = read_csv(path, header=None)
# split into input and output columns
X, y = df.values[:, :-1], df.values[:, -1]
# split into train and test datasets
X_train, X_test, y_train, y_test =
.
.
.

row2 =
np.array([[0.00632,18.00,2.310,0,0.5380,6.5750,65.20,4.0900,1,296.0,15.30,396.9
0,4.98]])
yhat = model.predict([row2])
print('Predicted: %.3f' % yhat)
```

Use the **Functional** model to develop a regression model using Boston Housing dataset, display

c) Squared sum error
d) Prediction results

6. Read Chapter 3: Improving the way neural networks learn at the following website

http://neuralnetworksanddeeplearning.com/chap3.html